Data Science Project: Titanic Survival Prediction

Project Description

This project focuses on analyzing the Titanic dataset and building a machine learning model to predict whether a passenger survived or not. It covers the complete data science pipeline from data preprocessing to visualization and prediction.

Objective

To understand data analysis, visualization, and machine learning by applying them to a real-world dataset.

Learning Outcomes & Skills

- Data Cleaning and Preprocessing using NumPy and Pandas
- Data Visualization using Seaborn and Matplotlib
- Feature Engineering
- Building Machine Learning Models
- Model Evaluation using Accuracy and Confusion Matrix

Steps Followed

1. Data Collection: Loaded Titanic dataset.
2. Data Cleaning: Handled missing values and removed unnecessary columns.
3. Data Visualization: Created graphs like bar plots, histograms, and heatmaps.
4. Feature Engineering: Created new features like FamilySize.
5. Model Building: Used Logistic Regression.
6. Evaluation: Checked accuracy and confusion matrix.

Features Used

- Age

- Sex
- Pclass
- Fare
- FamilySize

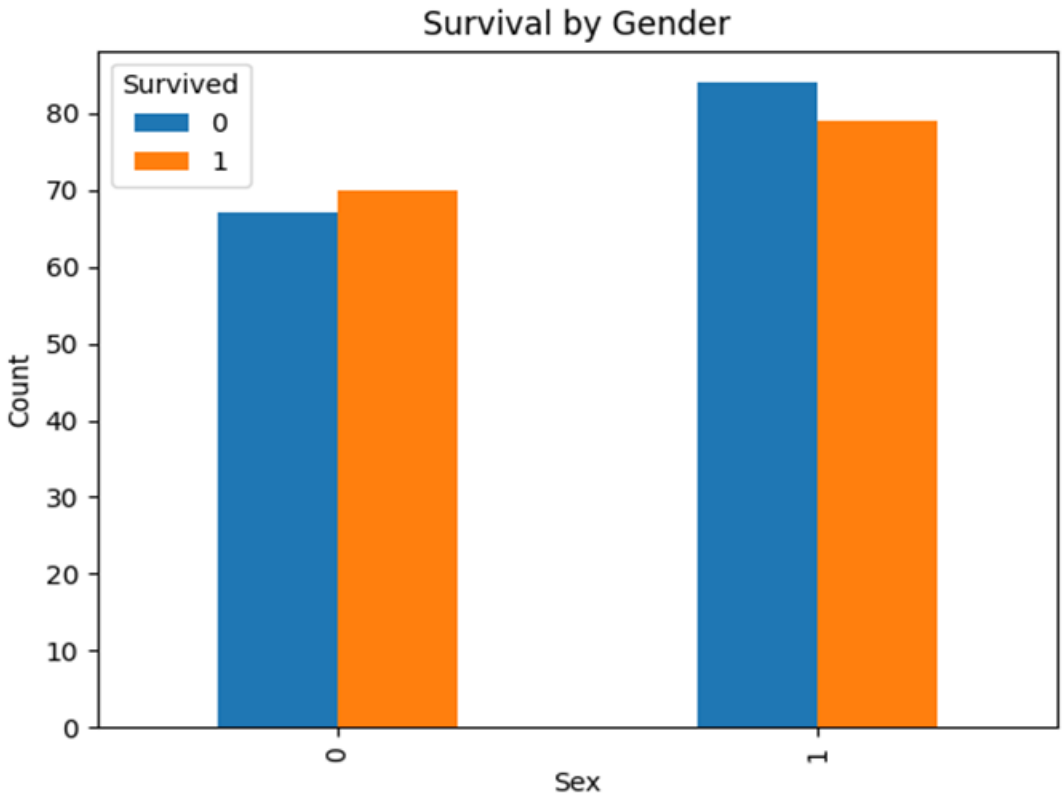
Machine Learning Model

Logistic Regression was used to classify whether a passenger survived or not based on input features.

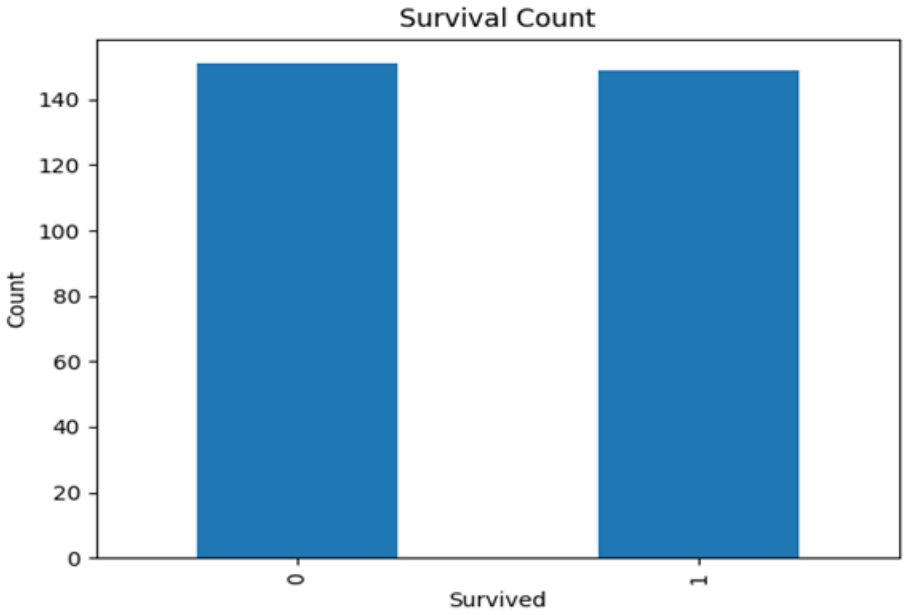
Results

The model achieved an accuracy of around 75–85%. The confusion matrix showed that most predictions were correct.

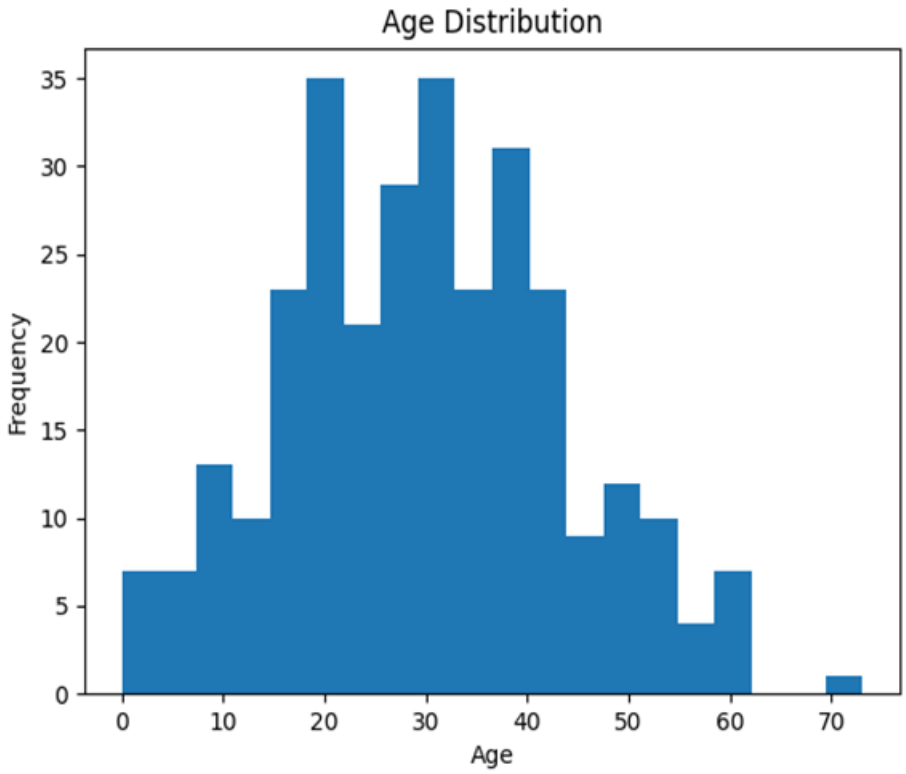
Graph for survival by gende



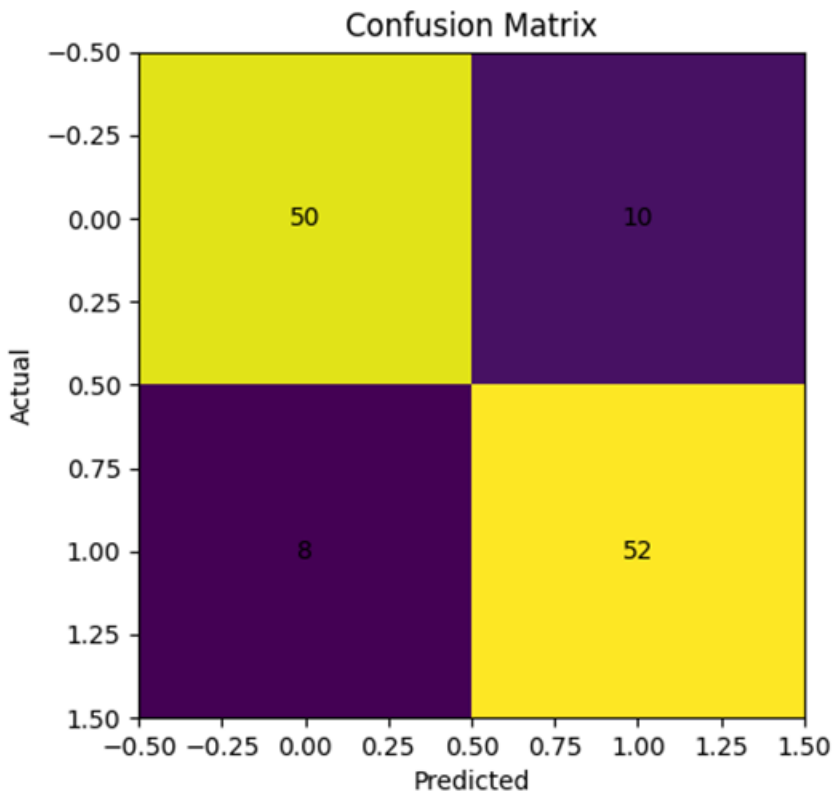
Graph for survival count



<u>Graph for age distribution</u>



Confusion matrix



Conclusion

This project demonstrates how data science techniques can be used to extract insights and make predictions. It provides a complete understanding of the data science workflow.